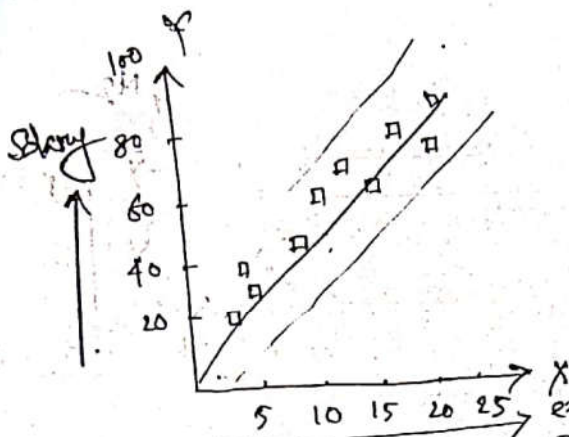


Prediction

x years experience	y salary (in \$1000s)	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
3	30	3-9.1 =	30-55.4 =			
8	57	8-9.1 =	57-55.4 =			
9	64	9-9.1 =	64-55.4 =			
13	72					
3	36					
6	43					
11	59					
21	90					
1	20					
16	83					
$\bar{x} = 9.1$	$\bar{y} = 55.4$					

- Question :
- plot the data. Do x and y seem to have a linear relationship
 - Use the least squares to find an equation for the predictions of an employee's salary fixed on the experience
 - Predict the salary of an employee who have 10 years experience.



Solution 01: From this figure we see that there exists a linear relationship.

Solution 02: For linear relationship we know

$$\hat{y} = w_0 + w_1 x \quad \text{--- (1)}$$

$$w_1 = \frac{\sum_{i=1}^D (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^D (x_i - \bar{x})^2}$$

Here, D = Total Number of Tuples
= 10

$$\therefore w_1 = \frac{(x_1 - \bar{x})(y_1 - \bar{y}) + \dots + (x_{10} - \bar{x})(y_{10} - \bar{y})}{(x_1 - \bar{x})^2 + \dots + (x_{10} - \bar{x})^2}$$

$$= \frac{(3-9.1)(30-55.4) + (8-9.1) + \dots + (16-9.1)(83-55.4)}{(3-9.1)^2 + (8-9.1)^2 + \dots + (16-9.1)^2} = 3.5$$

$$\therefore w_0 = \bar{y} - w_1 \bar{x} = 55.4 - 3.5 \times 9.1 = 23.6$$

Now the equation stands as follows
 $y = 23.6 + 3.5x$

Solution 03: Given $x = 10$ years
 $y = 23.6 + 3.5 \times 10 = 55$ (Ans.)

Related Math:

x Mid Term exam	y Final Exam
72	84
50	63
81	77
74	78
94	90
86	75
59	49
83	79
65	77
33	52
81	74
81	90

Ques: if $x = 86$ what will be the Final Exam Marks

$$y = mx + c$$